



Experimental AI Subseasonal-to-Seasonal (S2S) Forecast Guidance for India

Forecast Initialization: 26 August 2026

Forecast Period: 26 August–22 September 2026

Forecast Range: Weeks 1–4

Reference Dataset: ERA5 reanalysis/model climatology, 2002–2021

Forecast Product: IFS operational proxy, 100-member ensemble mean

Status: Experimental Research Guidance

Executive Summary

The 26 August 2026 S2S guidance indicates a transition from an active monsoon configuration during Week 1 toward a progressively weaker and more withdrawal-like circulation during Weeks 2–4. The experimental S2S guidance depicts a rainfall pattern centered primarily over east-central and eastern India during the first two forecast weeks, followed by a weakening and fragmentation of the rainfall anomaly during Weeks 3 and 4. In Week 1, the strongest positive rainfall anomaly is concentrated over Odisha, Chhattisgarh, adjoining eastern Madhya Pradesh, Jharkhand, and the northwestern Bay of Bengal. Much of western India and the southern peninsula are comparatively dry. During Week 2, positive rainfall anomaly mostly is confined to foot hills of Himalaya and to Northeast India keeping the rest of the country under negative anomalies. Week 1 represents the most spatially coherent continental wet signal in the forecast. This is consistent with the 850 hPa wind anomaly which shows the movement of monsoon trough close to foothills in week 2. Associated with this, the positive temperature anomaly is seen to be extended over north-western India and negative anomaly over foot hills and Northeast. The wet anomaly persists over only Northeast India during Week 3 and below-normal rainfall continues over much of the rest of India. By Week 4, the continental rainfall anomalies weaken considerably. Mostly dry anomalies become more evident across India except possibility of positive rainfall anomaly over southern Peninsula, the Himalayan foothills, and parts of Northeast India. Temperature anomalies remain dynamically consistent with the rainfall distribution. Central and eastern India are generally cooler than normal during Week 1. During Week 2, the cool anomaly shifts northward toward the Indo-Gangetic Plain, Himalayan foothills, and parts of Northeast India, while peninsular India becomes warmer than normal. During Weeks 3–4, warmer-than-normal conditions expand across most of central, northern, and peninsular India, while relatively cooler conditions remain localized near the Himalayan region and parts of Northeast India. By Week 4, anomalous easterly flow is evident across much of central and peninsular India, consistent with a substantially weakened southwest-monsoon circulation. The 850 hPa wind anomaly shows gradual weakening of monsoon flows with a cyclonic circulation at the south of west Bengal in week 1 and then gradual decrease of south-westerlies over Arabian sea and development of a ridge over central India by week 2 through to week 4, thus suggesting beginning of withdrawal of south-west monsoon. Week 4



anomalous easterlies are seen over southern peninsula. Forecast products are available at <https://s2s.scdids.com/>.

1. Methodology

The most prominent feature of the IFS initialization is a strong east-central Indian rainfall anomaly during Week 1, followed by a rapid north-eastward displacement of the wet signal during Week 2. Enhanced rainfall becomes concentrated along the Himalayan foothills and over Northeast India, while negative rainfall anomalies develop across most of central, western, and peninsular India. During Week 3, the continental wet signal is largely confined to Northeast India. By Week 4, weak positive anomalies emerge over parts of southern peninsular India, while rainfall anomalies elsewhere remain weak or negative. The low-level circulation changes from an active monsoon-like configuration during Week 1 to progressively weaker and more easterly anomalous flow over central and peninsular India during Weeks 2–4.

2. Rainfall Forecast

2.1 Week 1: 26 August–1 September 2026 The Week 1 rainfall forecast shows a pronounced wet anomaly over east-central India. The strongest positive departures are centered over Odisha and adjoining Chhattisgarh, eastern Madhya Pradesh, Jharkhand, and the northwestern Bay of Bengal. Locally substantial positive anomalies are also evident over the Bay of Bengal immediately southeast of the Odisha–West Bengal coast. The absolute rainfall field indicates that the principal continental rainfall axis extends from the west coast and central India eastward toward Odisha, West Bengal, and the northeastern states. The northeastern region also receives appreciable absolute rainfall, although the anomaly pattern is spatially mixed. Below-normal rainfall is forecast across much of western India, including portions of Gujarat, Rajasthan, Maharashtra, and the western peninsula. Negative anomalies also cover substantial parts of southern peninsular India and the southern Bay of Bengal. The circulation guidance shows anomalous westerly flow across the Arabian Sea and central peninsula, together with enhanced low-level flow extending toward the east-central Indian rainfall region. This configuration is consistent with continued moisture transport into central and eastern India and does not indicate a broad collapse of the monsoon circulation during Week 1.

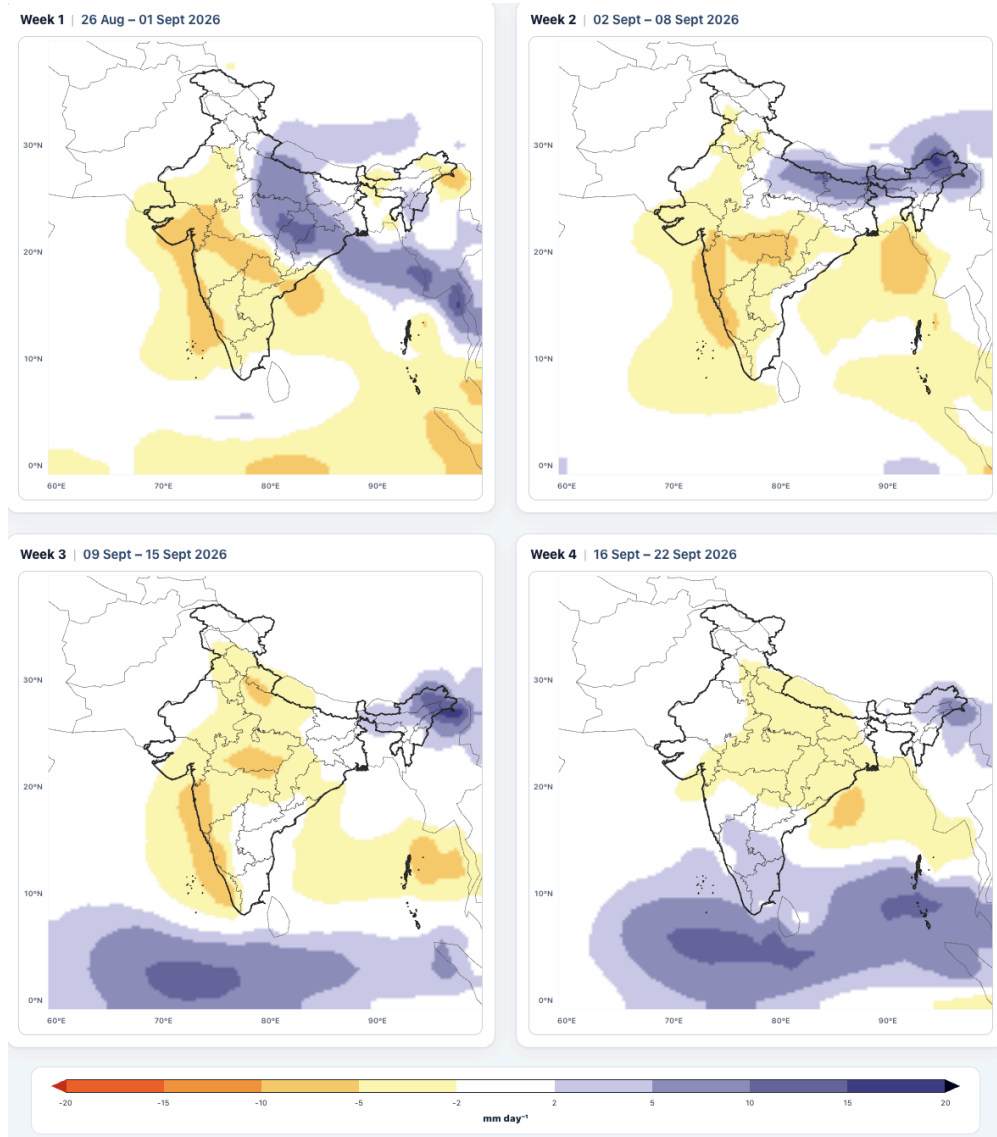


Figure 1. Weekly mean rainfall anomaly over India for Weeks 1–4, **IFS** operational proxy initialized 26 August 2026. Panels: (a) 26 August–1 September, (b) 2–8 September, (c) 9–15 September, and (d) 16–22 September 2026. Anomalies represent departures from the model’s typical rainfall for the corresponding season and forecast lead during 2002–2021.

2.2 Week 2: 2–8 September 2026 During Week 2, the positive rainfall anomaly becomes broader and shifts north-east and east-ward. During Week 2, the positive rainfall anomaly contracts and shifts north-eastward, becoming concentrated along the Himalayan foothills and over Northeast India. The rainfall anomaly over the Bay of Bengal broadly remains negative. Western India and much of the southern peninsula remain drier than normal. Negative anomalies are particularly evident over the western coastal and peninsular sectors and across



parts of the southern Bay of Bengal. Week 2 also shows development of an anomalous anti-cyclone over Arabian sea suggesting weakening of monsoon flow

2.3 Week 3: 9–15 September 2026 The Week 3 forecast retains a positive rainfall anomaly confined mostly over Northeast India, Below-normal rainfall remains widespread across western India and much of the peninsula. Negative anomalies are evident over Gujarat, Maharashtra, western Madhya Pradesh, and portions of southern India indicating possible withdrawal of monsoon flow. Enhanced rainfall is also forecast over the equatorial and southern Indian Ocean, but this oceanic signal is spatially separated from the principal continental rainfall anomaly. At 850 hPa, anomalous easterly flow continues across the Arabian Sea and central Indian latitudes.

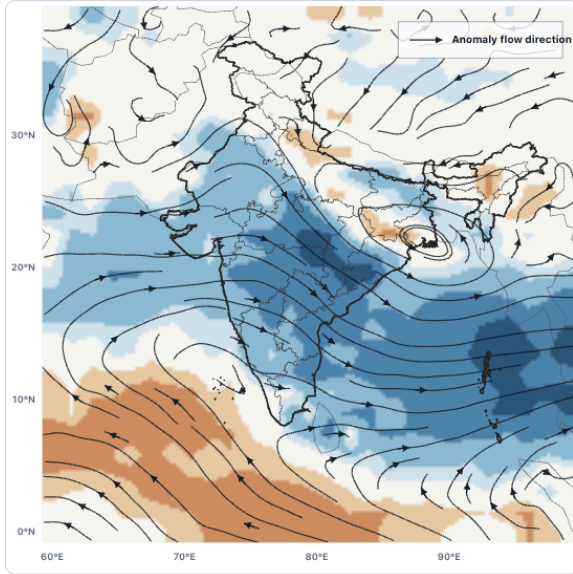
2.4 Week 4: 16–22 September 2026 By Week 4, the rainfall anomalies over India become substantially weaker and less organized. Modest positive anomalies remain over border portions of Northeast India, but the strong wet centers present during Weeks 1–2 are no longer evident. Negative rainfall anomalies develop across portions of northern India and the Himalayan foothill region. Weak negative anomalies are present over some areas of the western peninsula, while much of southern India is characterized by small positive departures from climatology. Positive rainfall anomalies strengthen over the southern Indian Ocean and parts of the southern Bay of Bengal. However, these features do not translate into a strong or widespread continental rainfall enhancement. The weaker and more fragmented Week 4 signal should be interpreted cautiously. At this lead time, the ensemble mean may smooth individual synoptic events, and forecast uncertainty is substantially greater than during Weeks 1–2.

3. Wind and Circulation Forecast

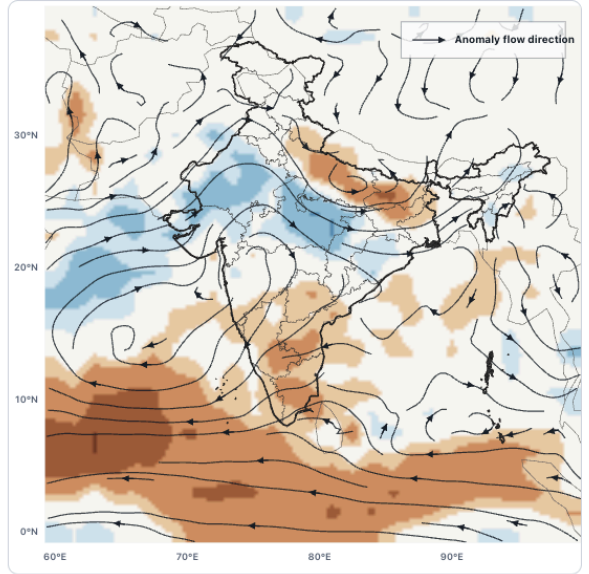
The 850-hPa wind and wind-anomaly forecasts provide dynamical context for the projected rainfall evolution. The anomaly maps show weekly wind-speed departures and anomalous vector components relative to the 2002–2021 reference period.

Week 1 During Week 1, enhanced low-level westerly anomalies extend from the Arabian Sea across peninsular and central India. The positive wind-speed anomaly is particularly pronounced across the central Arabian Sea, central peninsula, northern Bay of Bengal, and east-central Indian sector. The circulation is consistent with stronger-than-normal moisture transport across the central monsoon corridor. This supports the positive rainfall anomaly over Chhattisgarh, Odisha, eastern Madhya Pradesh, and adjoining regions. Anomalous easterly flow is present farther south over the equatorial Indian Ocean, but this feature is separated from the primary monsoon westerly belt and should not be interpreted as evidence of monsoon withdrawal over India.

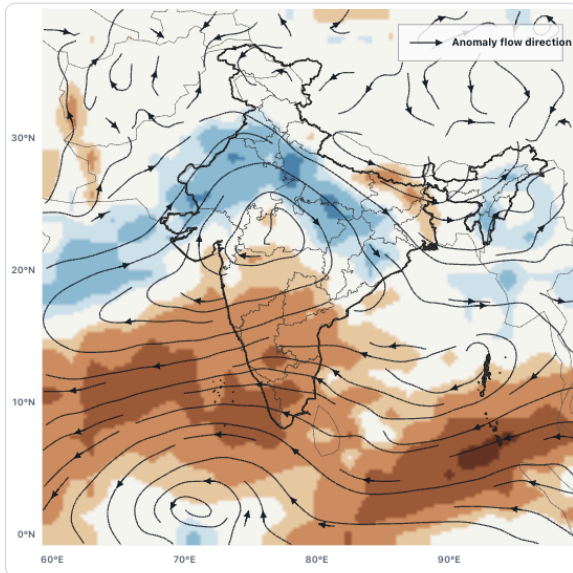
Week 1 | 26 Aug – 01 Sept 2026



Week 2 | 02 Sept – 08 Sept 2026



Week 3 | 09 Sept – 15 Sept 2026



Week 4 | 16 Sept – 22 Sept 2026

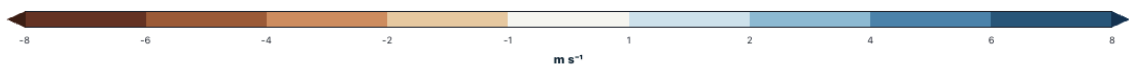
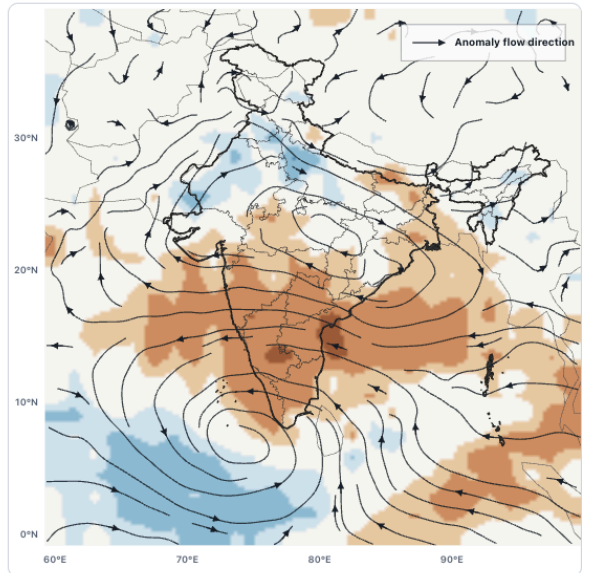


Figure 2. Weekly mean 850-hPa wind anomaly over India for Weeks 1–4, IFS operational proxy initialized 26 August 2026. Shading represents weekly wind-speed anomalies, and arrows represent anomalous U/V wind components relative to 2002–2021.



Week 2 The 850-hPa circulation undergoes a substantial reorganization compared with Week 1. The strong southwesterly/westerly anomaly over the central monsoon corridor weakens, while anomalous easterly flow becomes more prominent over the southern and central Indian Ocean–peninsular sector. The circulation pattern is therefore less supportive of sustained moisture transport into central India than during Week 1. The wet rainfall anomaly simultaneously becomes concentrated toward the Himalayan foothills and Northeast India. The strongest anomalous low-level circulation is positioned across approximately 15°–22°N, south of the Indo-Gangetic Plain. This configuration favors continued moisture transport into the central-eastern rainfall zone while leaving northwestern and southern peninsular India comparatively dry.

Week 3 During Week 3, anomalous easterly flow becomes established over much of the Arabian Sea, central India, and the Indian peninsula. The circulation is substantially less favorable for normal southwest-monsoon moisture transport into central India. This is dynamically consistent with the widespread negative rainfall anomalies across western, central, and peninsular India and the confinement of the positive rainfall signal to Northeast India.

Week 4 During Week 4, anomalous easterly flow remains evident across much of central and peninsular India, accompanied by an anticyclonic circulation anomaly over the broader central Indian region. The circulation is consistent with continued weakening of southwest-monsoon flow. However, because this is a Week 4 ensemble-mean forecast, the pattern should be described as withdrawal-like rather than evidence of formal monsoon withdrawal.

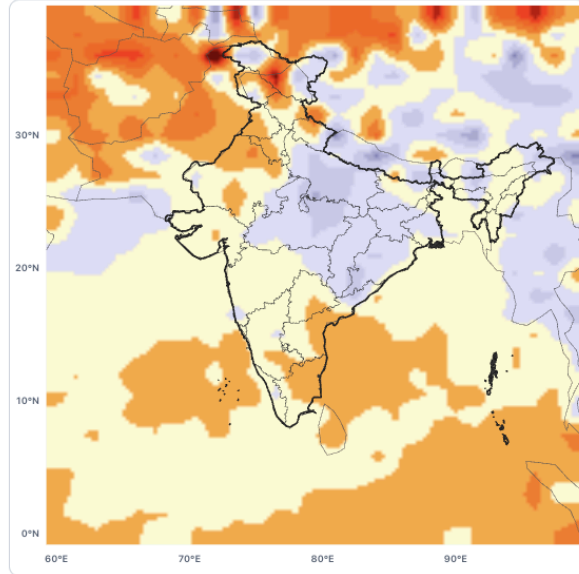
4. Temperature Forecast

4.1 Week 1: 26 August–1 September 2026 During Week 1, below-normal temperatures dominate central and eastern India. The most coherent cool anomaly extends across eastern Madhya Pradesh, Chhattisgarh, Odisha, Jharkhand, Bihar, and adjoining areas. This cool region corresponds closely with the enhanced rainfall anomaly and is physically consistent with increased rainfall, cloud cover, soil moisture, and reduced daytime surface heating. Warmer-than-normal conditions occur across northwestern and northern India, including portions of Rajasthan, Punjab, Haryana, and the western Himalayan region. Local positive anomalies are also present over southern interior peninsular India.

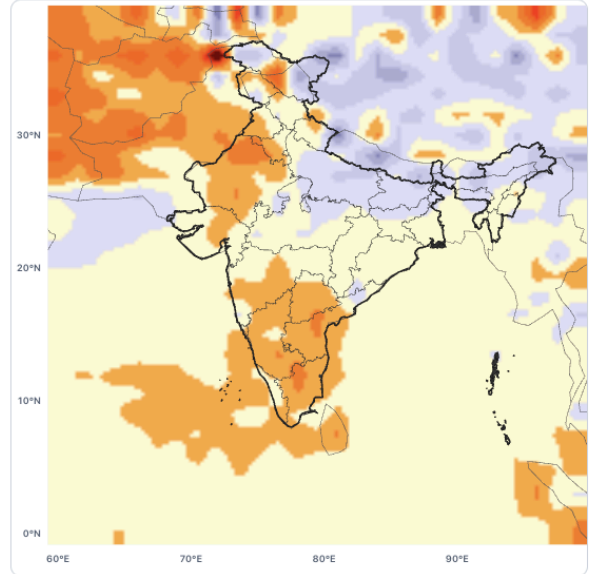


4.2 Week 2: 2–8 September 2026 During Week 2, below-normal temperatures become concentrated along the Indo-Gangetic Plain, Himalayan foothills, and parts of Northeast India. Central India is near normal to slightly warm, while positive temperature anomalies become evident across much of southern peninsular India. Strong warm anomalies continue over northwestern India and the region north of the Himalayas.

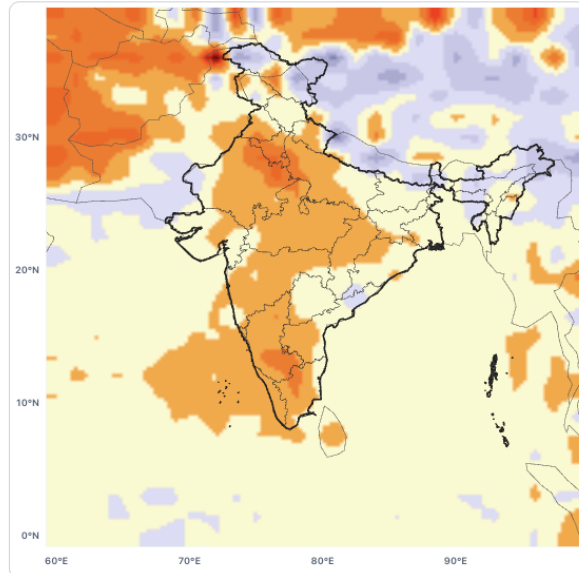
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Week 2 | 02 Sept – 08 Sept 2026



Week 3 | 09 Sept – 15 Sept 2026



Week 4 | 16 Sept – 22 Sept 2026

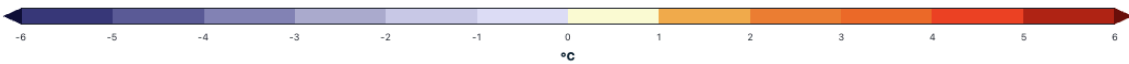
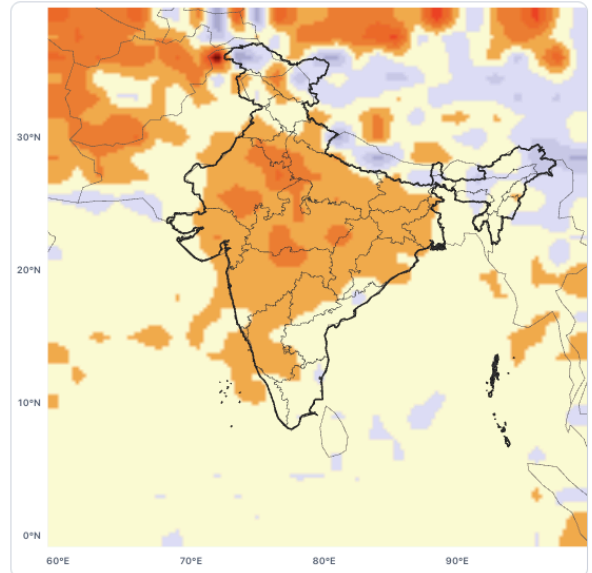


Figure 3. Weekly mean 2-m temperature anomaly over India for Weeks 1–4, IFS operational proxy initialized 26 August 2026. Anomalies are relative to the model's typical temperature for the corresponding season and forecast lead during 2002–2021.



4.3 Week 3: 9–15 September 2026 The temperature anomaly pattern changes substantially from Weeks 1–2. Positive temperature anomalies become widespread across central and peninsular India, while cooler conditions remain concentrated toward the Himalayan foothills and Northeast India. This warming occurs alongside continued negative rainfall anomalies across much of western and peninsular India and weakening low-level monsoon flow.

4.4 Week 4: 16–22 September 2026 Positive temperature anomalies become even more prominent across central, northern, eastern, and much of peninsular India, while cooler anomalies are relatively localized over Northeast India and parts of the southern oceanic region. The combination of warming and increasingly weak/negative rainfall anomalies is consistent with a seasonal transition away from the core southwest-monsoon regime.

5. Key S2S Signals

- **East-central rainfall enhancement during Weeks 1:** The principal wet signal extends across Odisha, Chhattisgarh, eastern Madhya Pradesh, Jharkhand, Bihar, and adjoining regions.
- **Negative anomaly during Week 2:** The positive rainfall anomaly becomes mostly confined to east and Northeast during 2–8 September, with the remaining part to be under negative rainfall anomaly.
- **Persistent dryness over western and peninsular India:** Much of Gujarat, Maharashtra, western Madhya Pradesh, and southern peninsular India remains near normal to drier than normal during Weeks 1–3.
- **Low-level circulation reversal:** Enhanced anomalous westerly flow supports an active monsoon configuration during Week 1. From Week 2 onward, the westerly anomalies weaken and anomalous easterly flow becomes increasingly prominent over central and peninsular India.
- **Cool eastern India:** Negative temperature anomalies weeks 1-2 coincide with the principal rainfall-enhancement region throughout much of the forecast.
- **Warm northwestern and northern India:** Positive temperature anomalies persist over the comparatively dry northwest, with locally strong warming along the Himalayan region.
- **Withdrawal-like signal:** The Week 1 circulation remains monsoon-active. During Weeks 2–4, weakening westerlies, expanding easterly anomalies, and increasingly dry and warm conditions over central and western India indicate a possible transition toward monsoon withdrawal. The forecast does not establish the formal timing of withdrawal.

6. Interpretation



The 26 August 2026 IFS-initialized guidance depicts an active monsoon configuration during Week 1, followed by a pronounced weakening and north-eastward displacement of the continental rainfall signal. During Week 1, enhanced rainfall is concentrated over Odisha, Chhattisgarh, eastern Madhya Pradesh, Jharkhand, and adjoining regions, supported by an active low-level monsoon circulation. In Week 2, the wet anomaly contracts toward the Himalayan foothills and Northeast India, while most of central, western, and peninsular India becomes drier than normal. By Week 3, positive continental rainfall anomalies are largely confined to Northeast India. Week 4 shows weak positive rainfall anomalies over parts of southern peninsular India and Northeast India, but no widespread continental wet signal. The circulation evolution is consistent with this rainfall transition. Stronger monsoon-like flow during Week 1 gives way to weaker westerlies and increasingly prominent anomalous easterly flow over central and peninsular India during Weeks 2–4. Temperatures similarly evolve from cooler-than-normal conditions over central and eastern India during Week 1 to widespread warmer-than-normal conditions across much of India during Weeks 3–4. Together, these signals indicate a progressive weakening of the southwest-monsoon circulation and the emergence of withdrawal-like conditions. However, the forecast does not provide sufficient evidence to determine the formal timing of monsoon withdrawal. Confidence is highest for Week 1, moderate for Week 2, and substantially lower for Weeks 3–4.